Examples Radar Design Trades
EEL 5557 S26

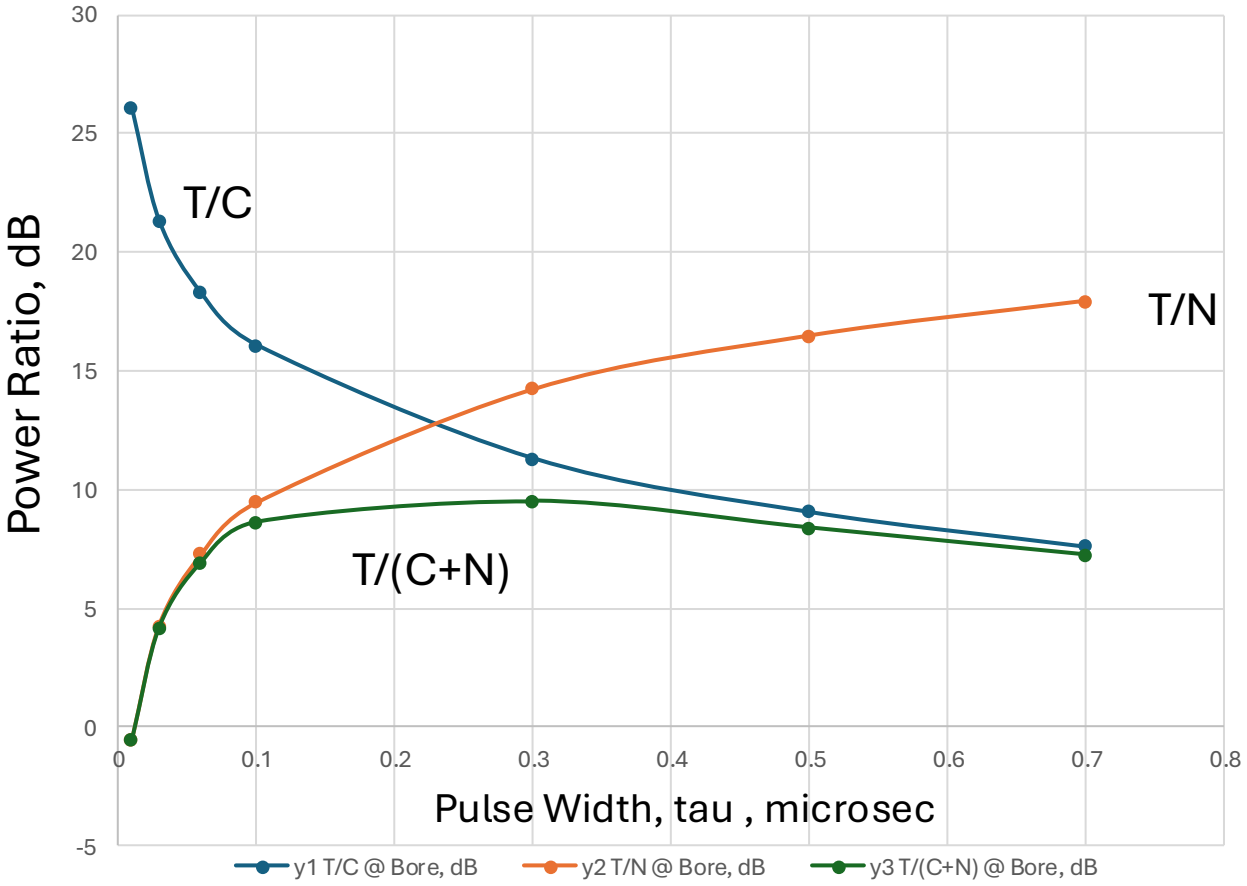
Examples of Trades for S26

- This presentation gives examples of typical results that you should obtain during your Radar design project
- They are meant to be examples that you should obtain, but NOT a recommended design
- Dr J has conducted numerous trade studies that indicate that there are many spec compliant designs possible for many combinations of frequencies, altitudes and boresight pointing angles
  - Once you have a spec compliant design, you should optimize the design to meet (TBD) selected goals
    - All goals are equally weighted
    - It is highly unlikely that all goals can be simultaneously satisfied
  - Your Radar Design Proposal should present several (3 or more) spec-compliant designs and your rational for your final design selection

Signal to Noise Expectations
(with varying Pulse Width, tau)

All Constant

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer

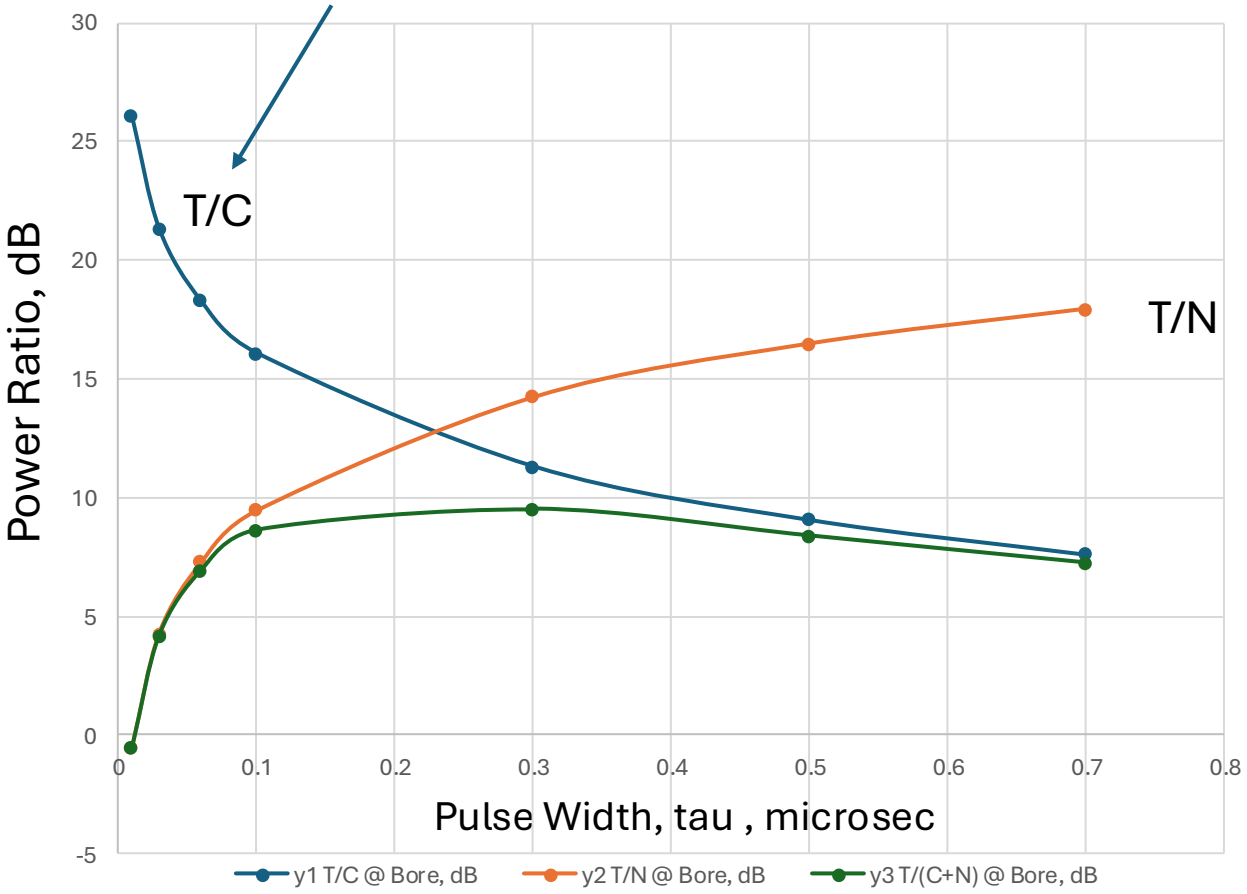


Signal to Noise Expectations
(with varying Pulse Width, tau)

All Constant

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer

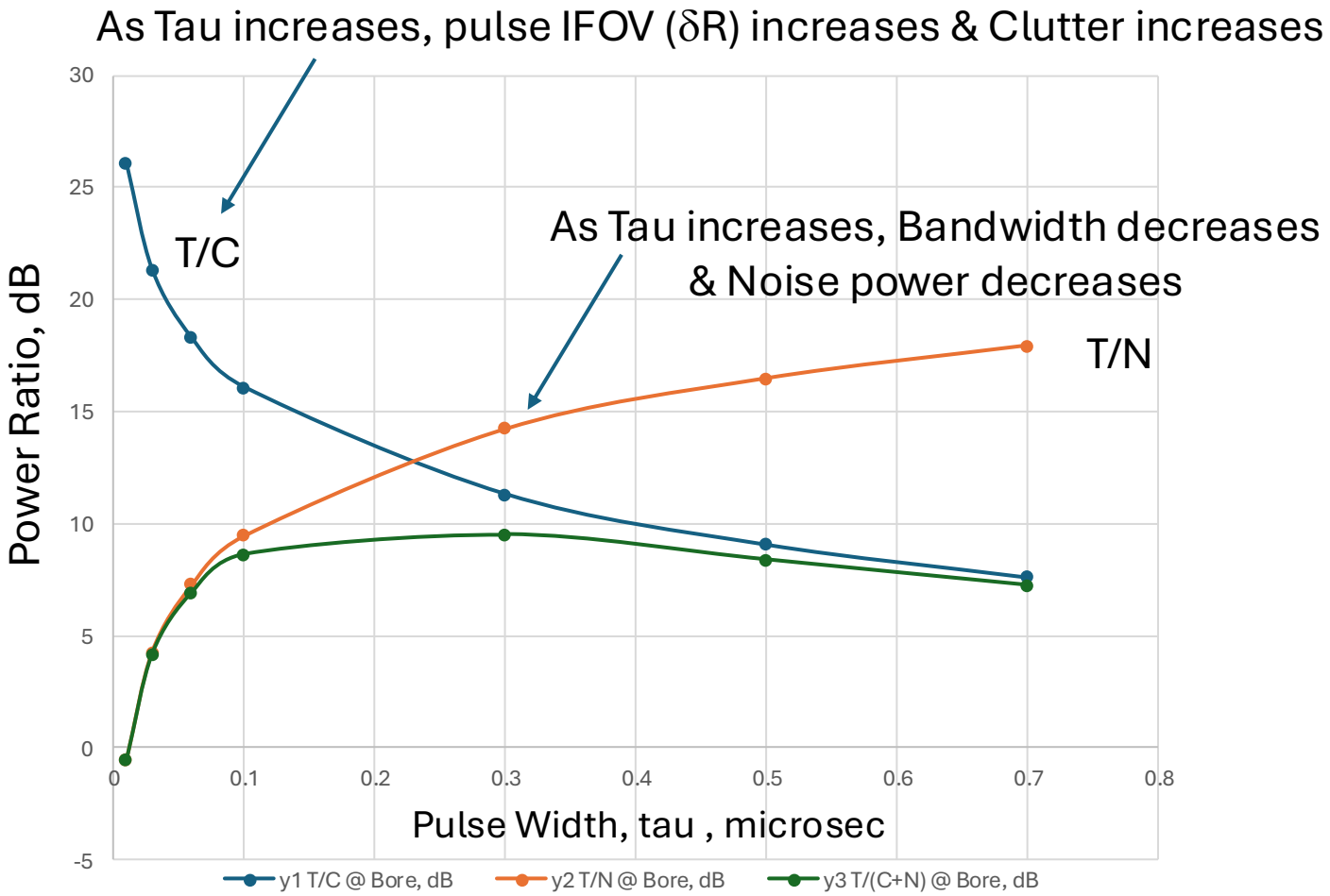
As Tau increases, pulse IFOV (δR) increases & Clutter increases



Signal to Noise Expectations (with varying Pulse Width, tau)

All Constant

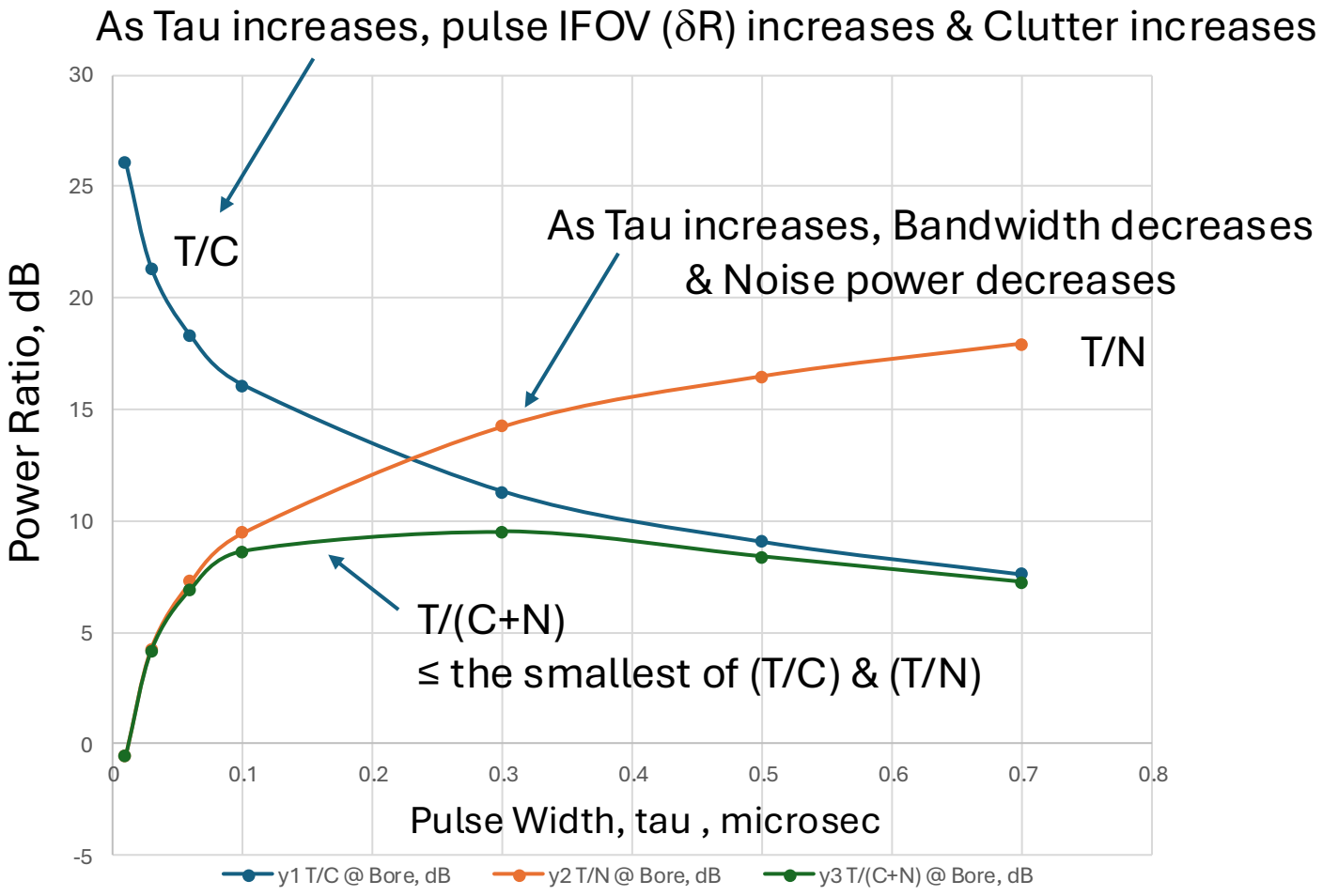
Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer



Signal to Noise Expectations (with varying Pulse Width, tau)

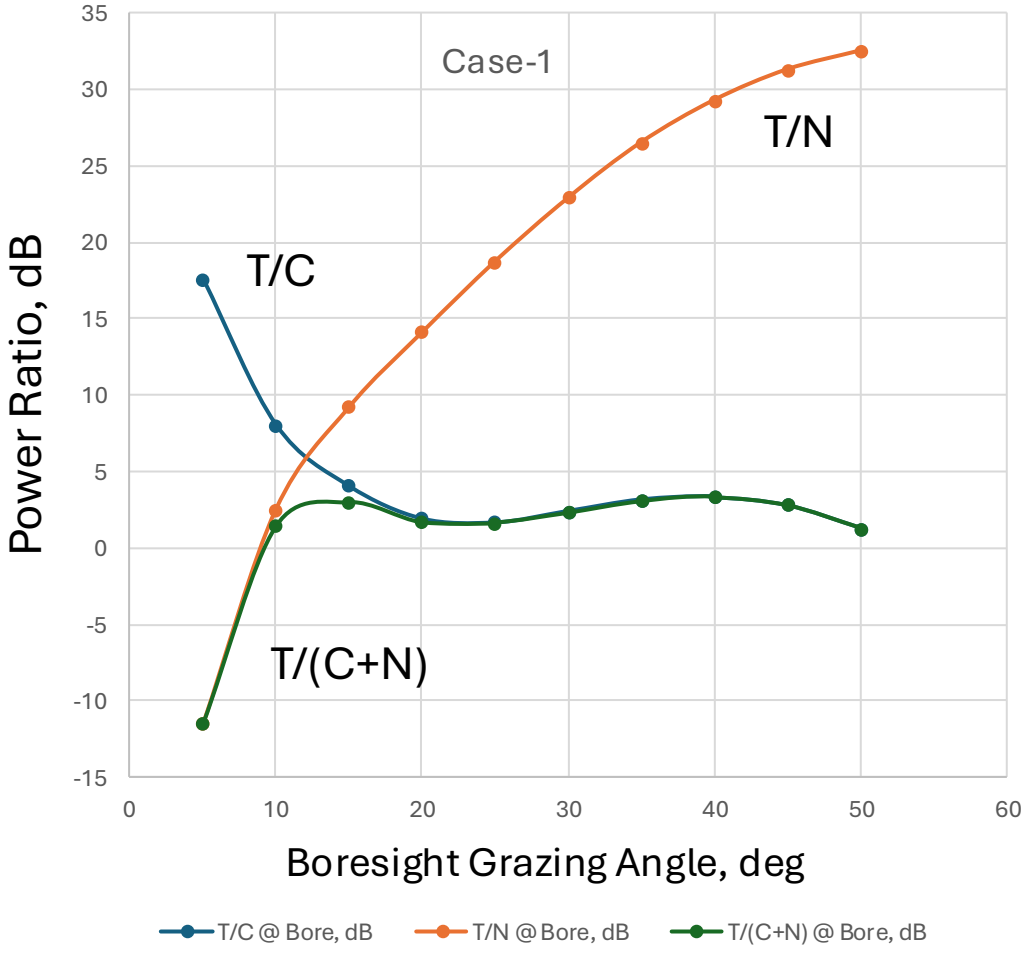
All Constant

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant_Bore_grazing_Ang		
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmt power		W
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer

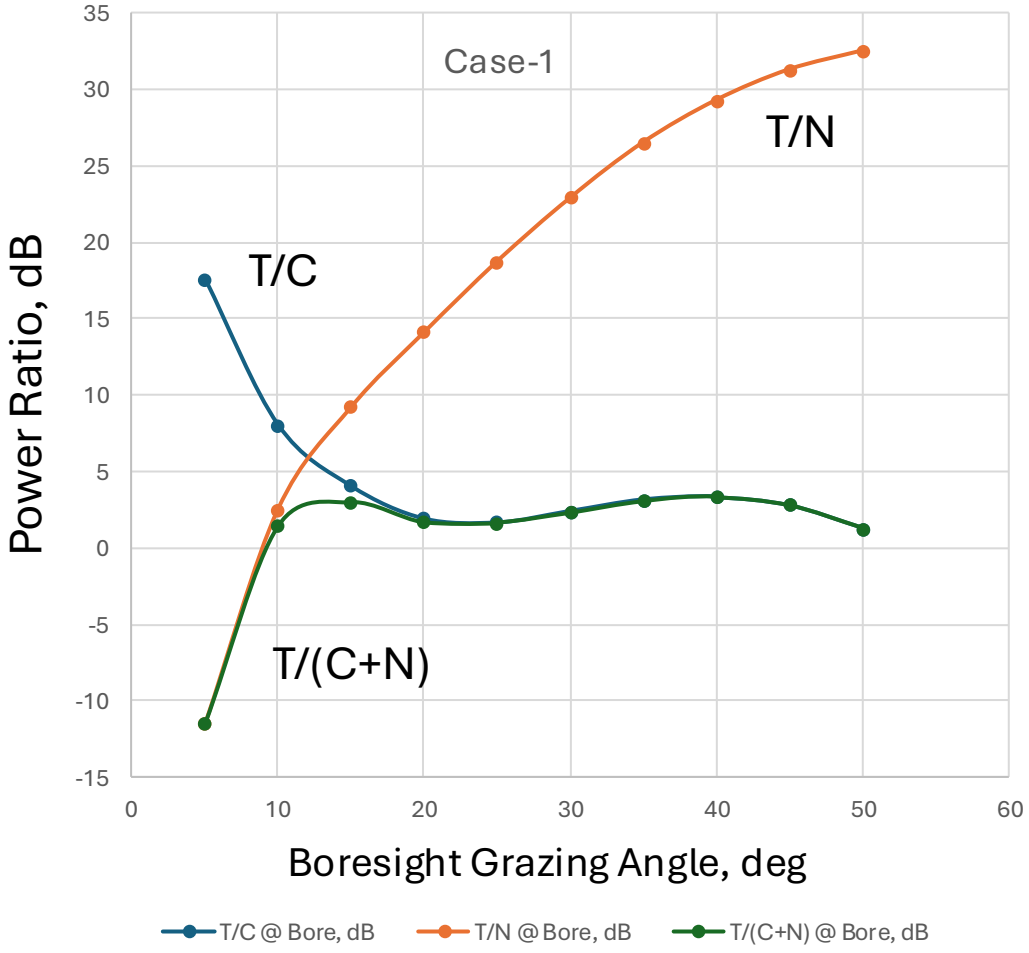


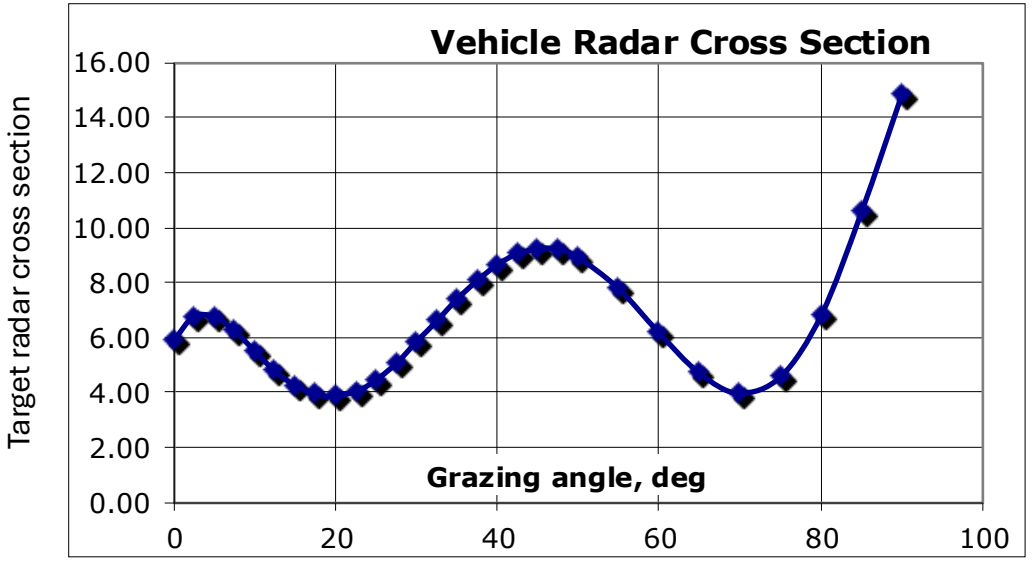
Signal to Noise Expectations

(with varying Boresight Grazing Angle)

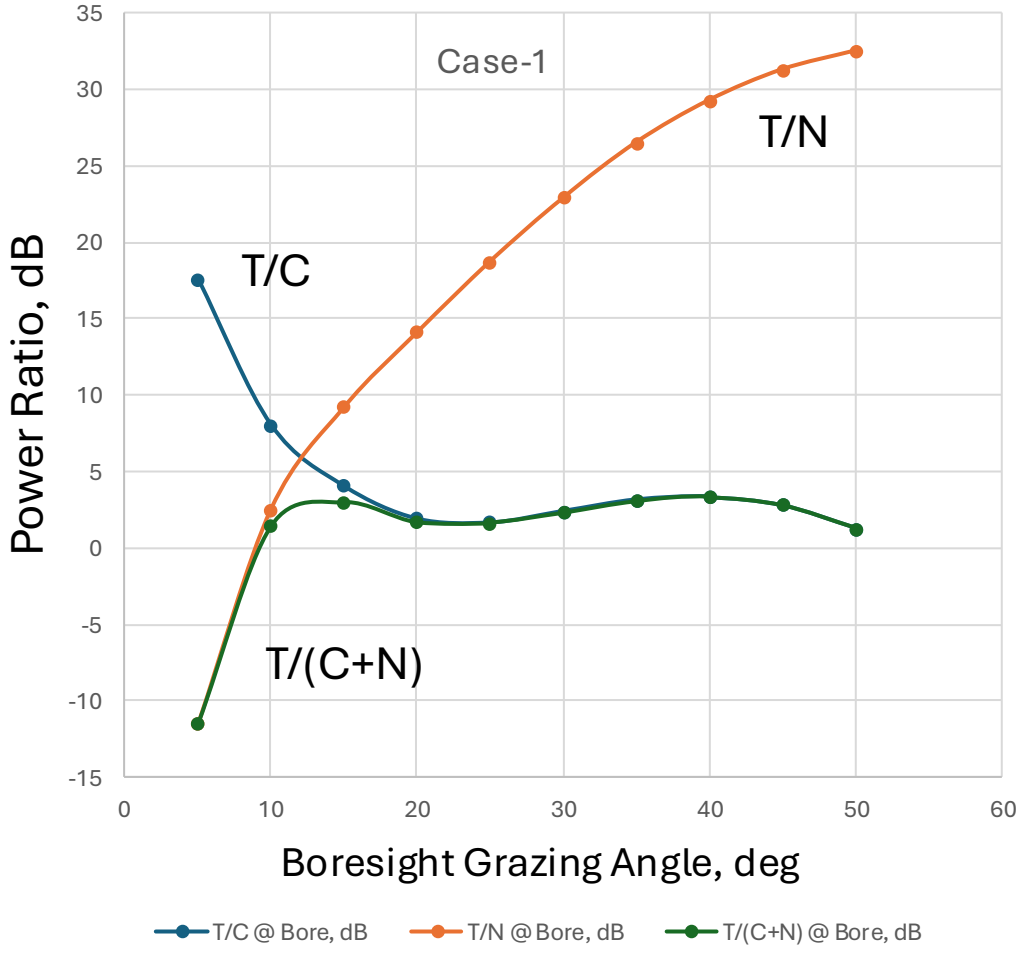


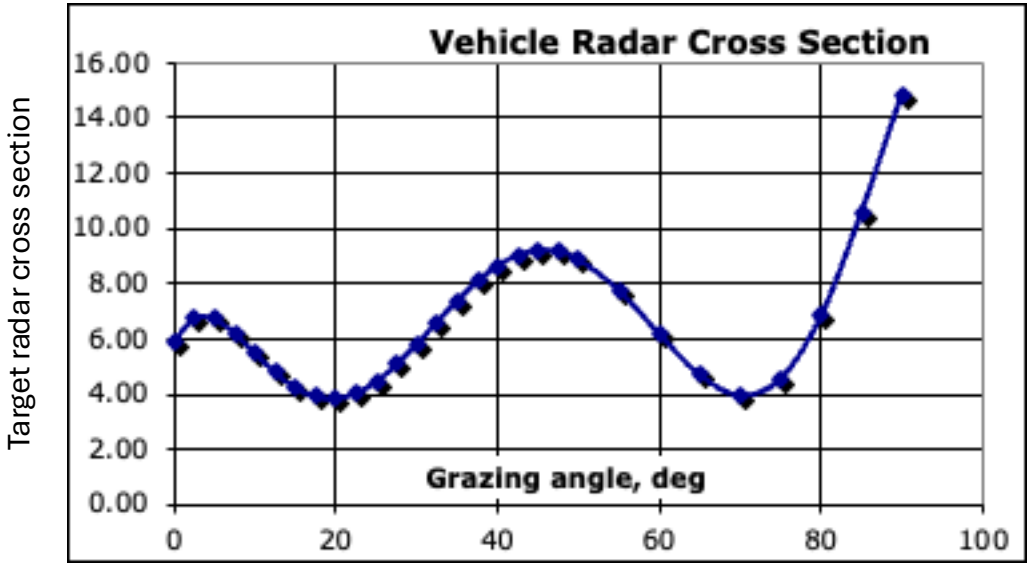
Signal to Noise Expectations
(with varying Boresight Grazing Angle)

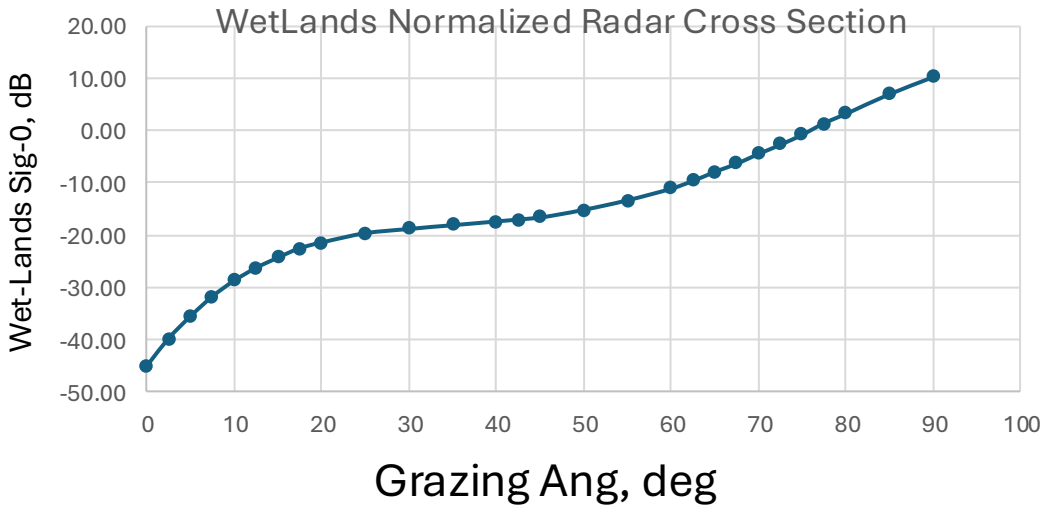




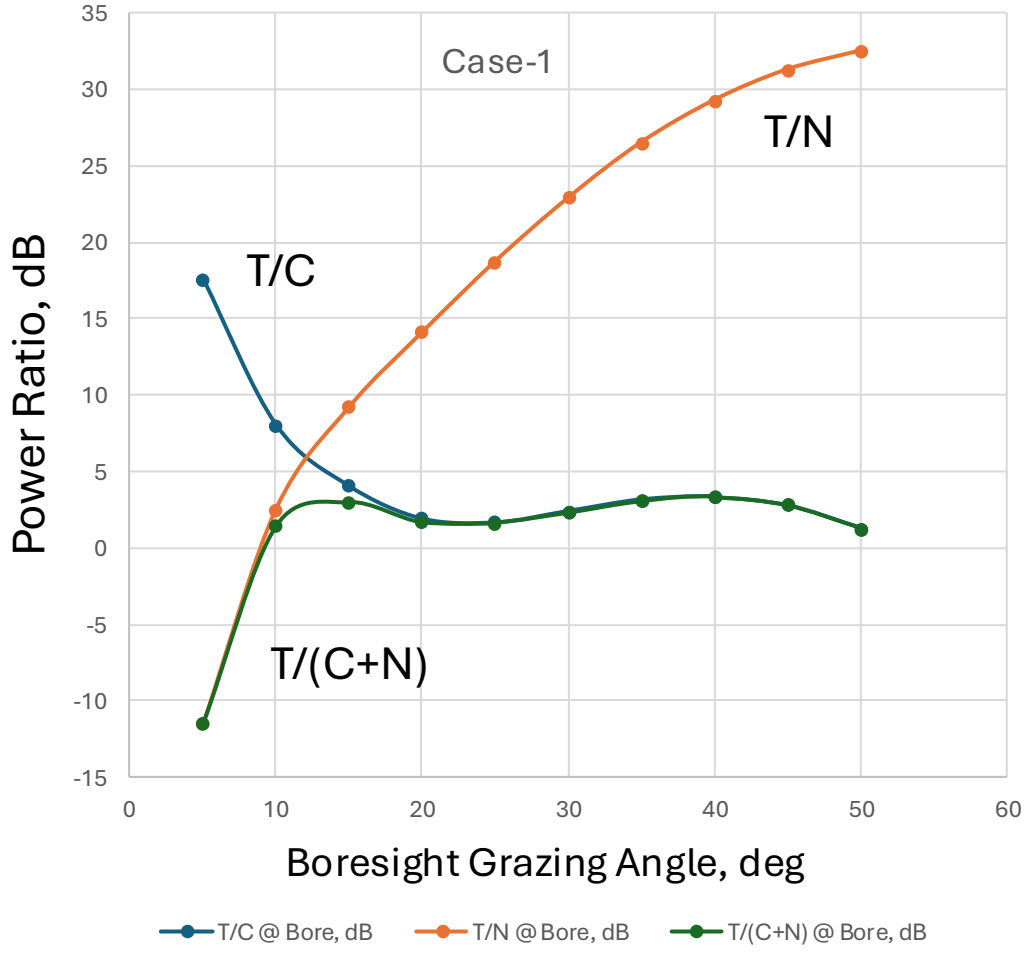
Signal to Noise Expectations
(with varying Boresight Grazing Angle)

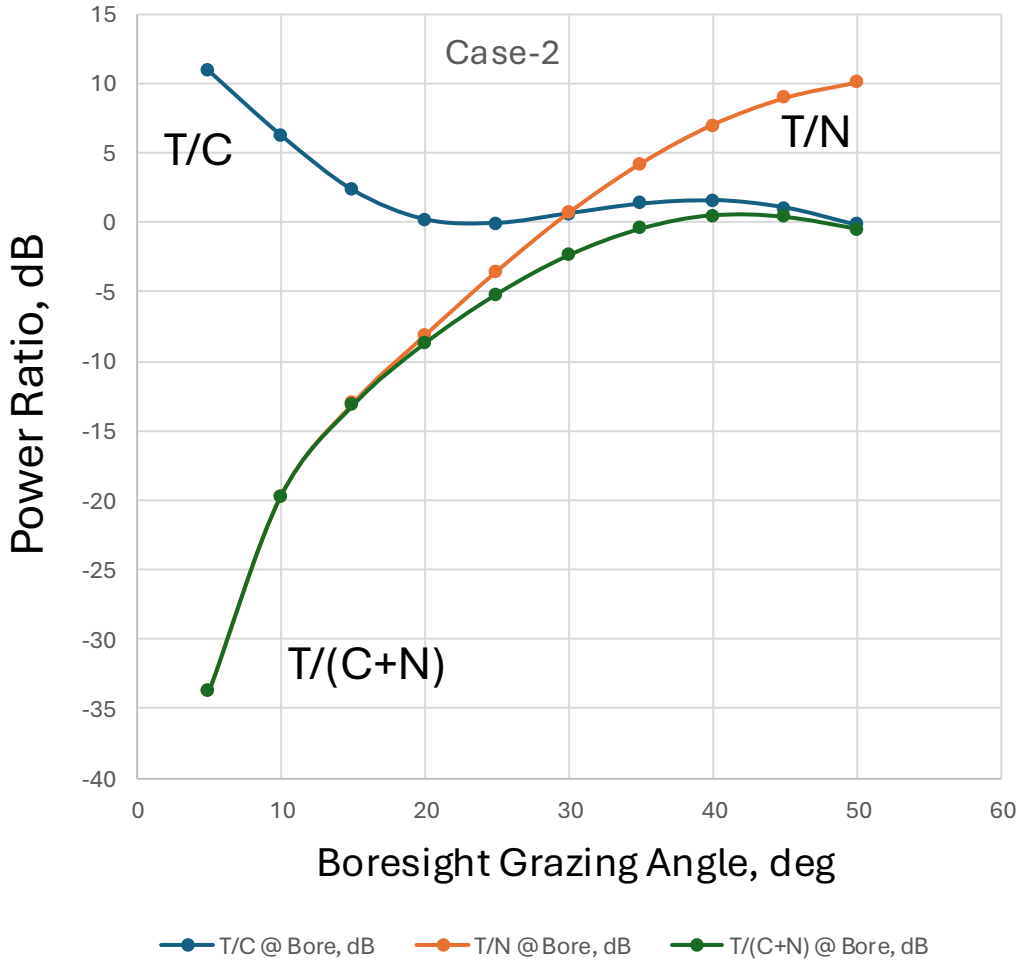




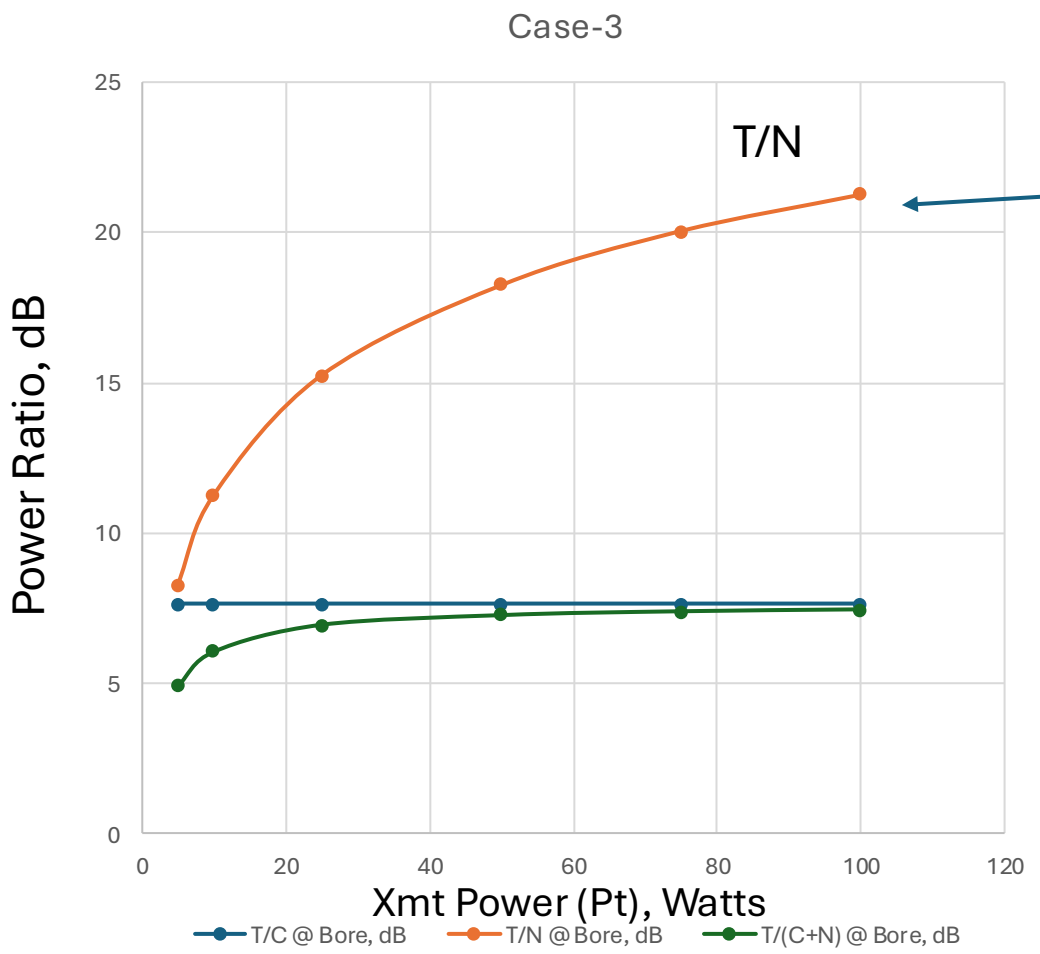


Signal to Noise Expectations
(with varying Boresight Grazing Angle)



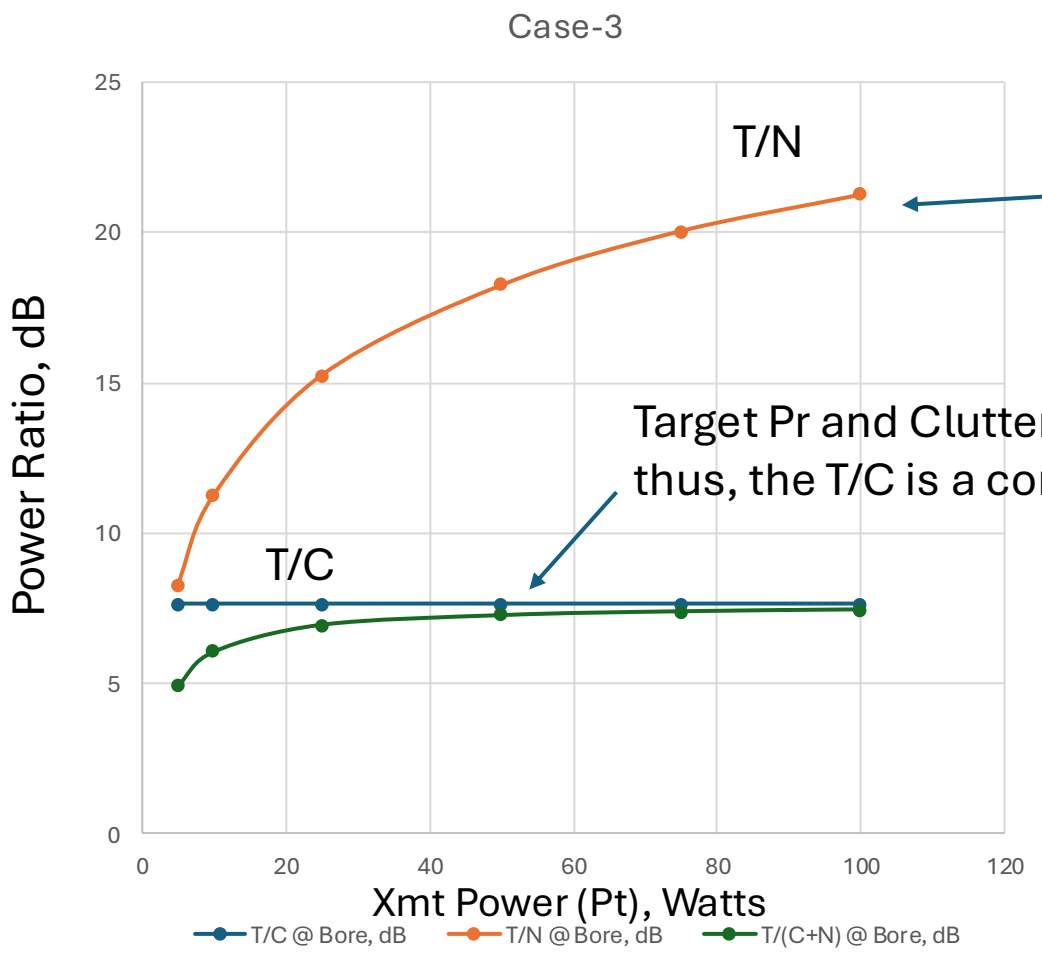


Signal to Noise Expectations (with varying Xmt Power, Pt)



Radar X-factor increases with Xmt Pwr (Pt); thus, Target Pr increases with Pt

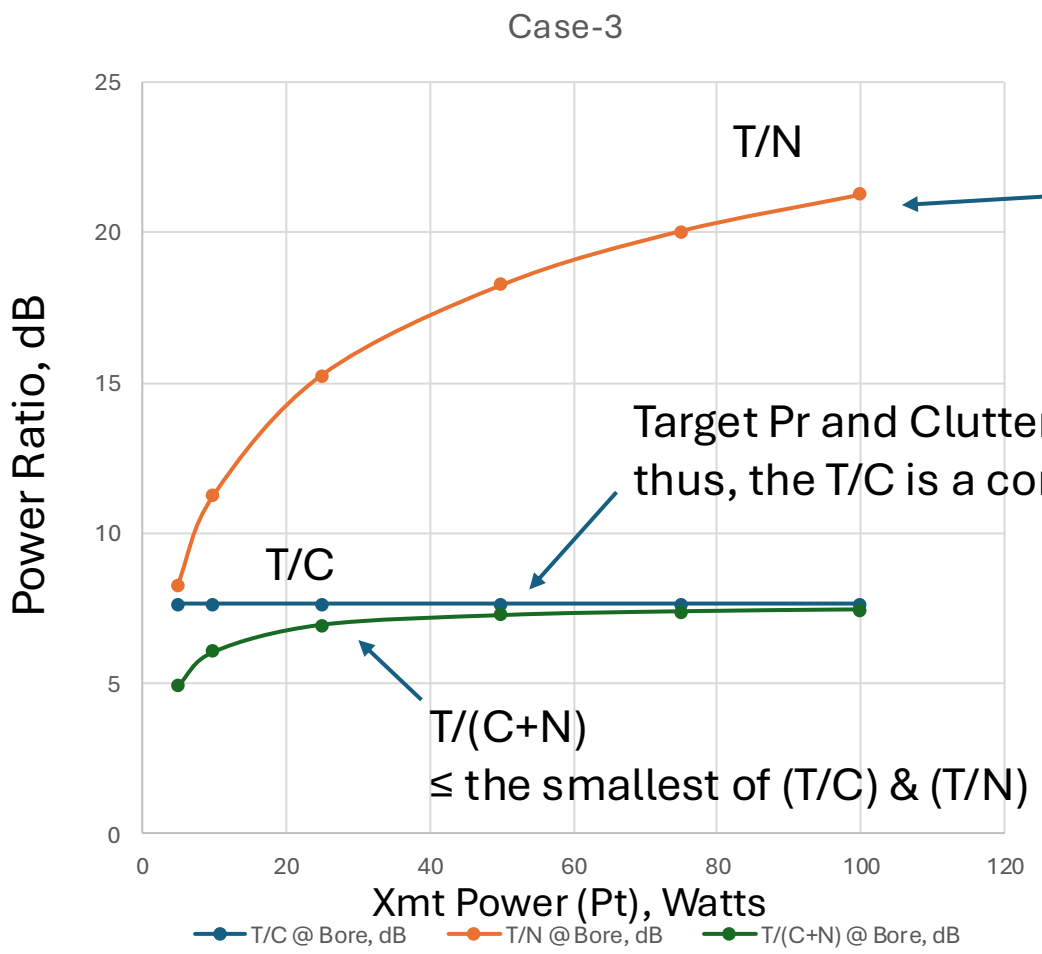
Signal to Noise Expectations (with varying Xmt Power, Pt)



Radar X-factor increases with Xmt Pwr (Pt); thus, Target Pr increases with Pt

Target Pr and Clutter Pr share a constant X-factor; thus, the T/C is a constant

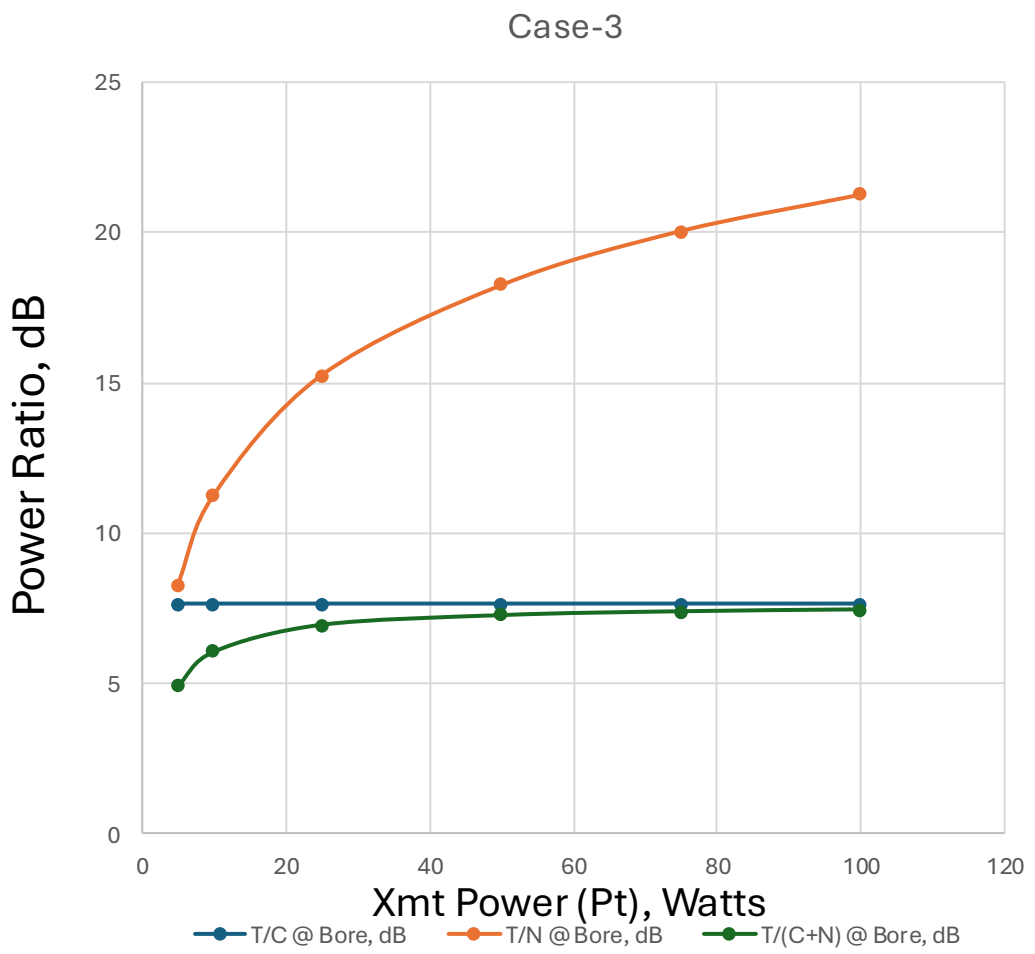
Signal to Noise Expectations (with varying Xmt Power, Pt)

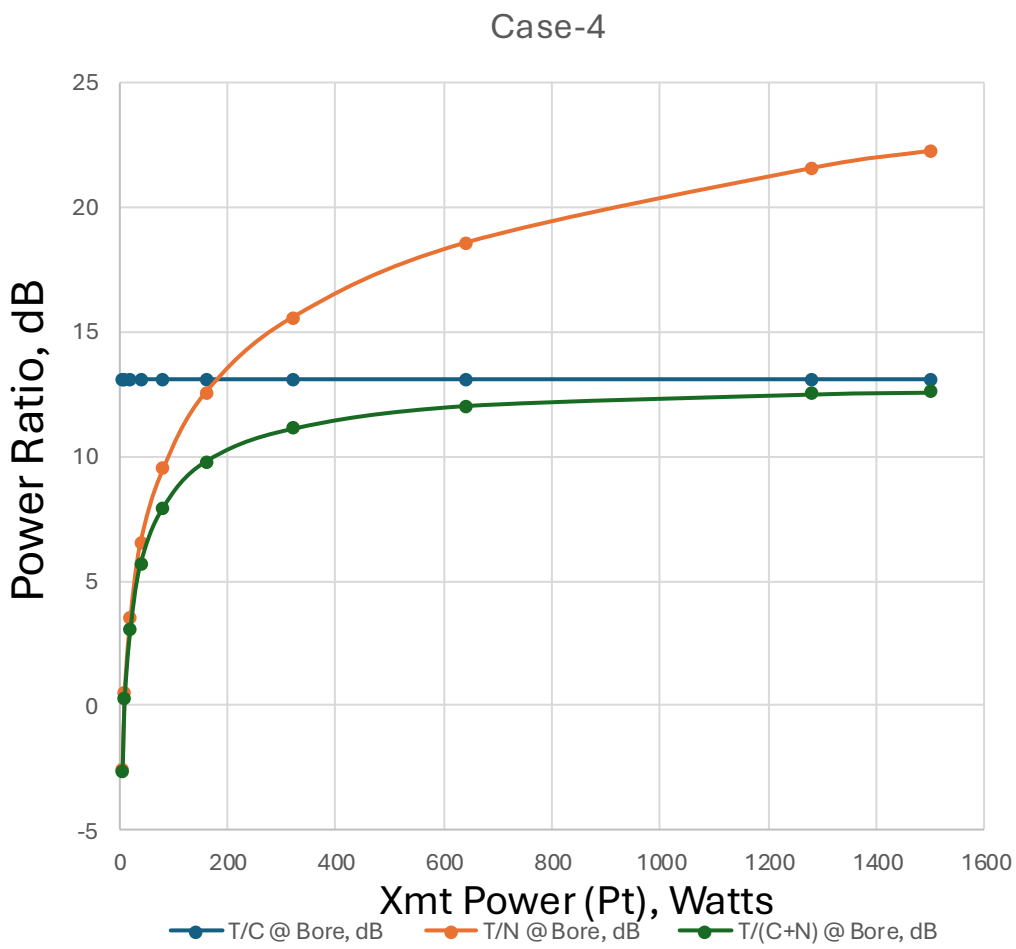


Radar X-factor increases with Xmt Pwr (Pt); thus, Target Pr increases with Pt

Target Pr and Clutter Pr share a constant X-factor; thus, the T/C is a constant

Signal to Noise Expectations (with varying Xmt Power, Pt)





Comparison of Goals Achieved for Freq = 150 GHz

Case-A

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.500	meter
Ant dia- azimuth		meter
Ant spin rate	80	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	450.00	W
Xmt pulse width		microsec
PRF		Hz
<Tfa>	8.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.50	0.10	71%
Ant Az dia		0.10	
Target tracking, # spins	3	10	0%
Ant Spin rate	80	30	67%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	450	10.0	70%
Swath width		5.00E+04	
Total goals score			lowest

Comparison of Goals Achieved for Freq = 150 Ghz

Case-B

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.300	meter
Ant dia- azimuth		meter
Ant spin rate	50	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	550.00	W
Xmt pulse width		microsec
PRF		Hz
<Tfa>	12.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.30	0.10	86%
Ant Az dia		0.10	
Target tracking, # spins	3	10	0%
Ant Spin rate	50	30	87%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	550	10.0	63%
Swath width			
Total goals score			middel

Comparison of Goals Achieved for Freq = 150 Ghz

Case-C

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	1.50E+11	Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation	0.200	meter
Ant dia- azimuth		meter
Ant spin rate	35	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	825.00	W
Xmt pulse width		microsec
PRF		Hz
<Tfa>	18.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.20	0.10	93%
Ant Az dia		0.10	
Target tracking, # spins	3	10	0%
Ant Spin rate	35	30	97%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	825	10.0	45%
Swath width		5.00E+04	
Total goals score			highest

Comparison of Goals Achieved for Freq = 150 Ghz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C
Radar Freq	1.50E+11	1.50E+11	1.50E+11
UAV Altitude			
Ant.boresight grazing ang			
Ant dia- elevation	0.500	0.300	0.200

Decreasing diameter, increasing IFOV

Comparison of Goals Achieved for Freq = 150 Ghz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C
Radar Freq	1.50E+11	1.50E+11	1.50E+11
UAV Altitude			
Ant.boresight grazing ang			
Ant dia- elevation	0.500	0.300	0.200
Ant dia- azimuth			
Ant spin rate	80	50	35

Decreasing diameter, increasing IFOV

Reduced spin-rate for tracking target

Comparison of Goals Achieved for Freq = 150 Ghz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C
Radar Freq	1.50E+11	1.50E+11	1.50E+11
UAV Altitude			
Ant.boresight grazing ang			
Ant dia- elevation	0.500	0.300	0.200
Ant dia- azimuth			
Ant spin rate	80	50	35
Antenna Azimuth +/- from fore-look			
Xmtr Pwr	450.00	550.00	825.00

Decreasing diameter, increasing IFOV

Reduced spin-rate for tracking target

Increased Pt to achieve req T/(C+N) and/or T/N

Comparison of Goals Achieved for Freq = 150 Ghz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C
Radar Freq	1.50E+11	1.50E+11	1.50E+11
UAV Altitude			
Ant.boresight grazing ang			
Ant dia- elevation	0.500	0.300	0.200
Ant dia- azimuth			
Ant spin rate	80	50	35
Antenna Azimuth +/- from fore-look			
Xmtr Pwr	450.00	550.00	825.00
Xmt pulse width			
PRF			
<Tfa>	8.00	12.00	18.00

Decreasing diameter, increasing IFOV

Reduced spin-rate of tracking target

Increased Pt to achieve req T/(C+N) and/or T/N

Tracks spin rate <Tfa>

Comparison of Goals Achieved for Freq = 150 Ghz

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C
Radar Freq	1.50E+11	1.50E+11	1.50E+11
UAV Altitude			
Ant.boresight grazing ang			
Ant dia- elevation	0.500	0.300	0.200
Ant dia- azimuth			
Ant spin rate	80	50	35
Antenna Azimuth +/- from fore-look			
Xmtr Pwr	450.00	550.00	825.00
Xmt pulse width			
PRF			
<Tfa>	8.00	12.00	18.00
Design capability: # scans on target = N	3	3	3
PCR	1.00	1.00	1.00
Detection to Doppler (Time/scan)*	1	1	1

Decreasing diameter, increasing IFOV

Reduced spin-rate of tracking target

Increased Pt to achieve req T/(C+N) and/or T/N

Tracks spin rate <Tfa>

Total goals score	lowest	middel	highest

% Achieved Goals

37 GHz Cases

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Case-A

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	6.0	deg
Ant dia- elevation	1.000	meter
Ant dia- azimuth	1.40	meter
Ant spin rate	15	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	850.00	W
Xmt pulse width	0.1500	microsec
PRF	85,000	Hz
<Tfa>	40.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	1.00	0.10	36%
Ant Az dia	1.40	0.10	7%
Target tracking, # spins	3	10	0%
Ant Spin rate	15	30	100%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	850	10.0	43%
Swath width		5.00E+04	
Total goals score			Rank-2 (lower)

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Case-B

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	12.0	deg
Ant dia- elevation	0.900	meter
Ant dia- azimuth	1.00	meter
Ant spin rate	50	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	200.00	W
Xmt pulse width	0.0800	microsec
PRF	300,000	Hz
<Tfa>	12.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.90	0.10	43%
Ant Az dia	1.00	0.10	36%
Target tracking, # spins	3	10	0%
Ant Spin rate	50	30	87%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	200	10.0	87%
Swath width		5.00E+04	
Total goals score			Rank-1 (highest)

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Case-C

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	18.0	deg
Ant dia- elevation	0.870	meter
Ant dia- azimuth	1.50	meter
Ant spin rate	100	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	100.00	W
Xmt pulse width	0.0600	microsec
PRF	500,000	Hz
<Tfa>	12.00	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.87	0.10	45%
Ant Az dia	1.50	0.10	0%
Target tracking, # spins	3	10	0%
Ant Spin rate	100	30	53%
Warning time		60	100%
Doppler precision goal (worst-case)		50.0	
Transmit power	100	10.0	93%
Swath width		5.00E+04	
Total goals score			Rank-3 (lower)

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Case-D

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	24.0	deg
Ant dia- elevation	0.570	meter
Ant dia- azimuth	1.50	meter
Ant spin rate	115	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	60.00	W
Xmt pulse width	0.0380	microsec
PRF	500,000	Hz
<Tfa>	5.50	sec
Design capability: # scans on target = N	3	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.57	0.10	66%
Ant Az dia	1.50	0.10	0%
Target tracking, # spins	3	10	0%
Ant Spin rate	115	30	43%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	60	10.0	96%
Swath width		5.00E+04	
Total goals score			Rank-4 (lowest)

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0

Increasing grazing angle

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570

Increasing El beam width

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115

Increasing spin rate

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	850.00	200.00	100.00	60.00

decreasing Pt

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	850.00	200.00	100.00	60.00
Xmt pulse width	0.1500	0.0800	0.0600	0.0380

decreasing tau

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	850.00	200.00	100.00	60.00
Xmt pulse width	0.1500	0.0800	0.0600	0.0380
PRF	85,000	300,000	500,000	500,000

increasing PRF

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	850.00	200.00	100.00	60.00
Xmt pulse width	0.1500	0.0800	0.0600	0.0380
PRF	85,000	300,000	500,000	500,000
<Tfa>	40.00	12.00	12.00	5.50

Tracks spin rate <Tfa>

Comparison of Goals Achieved for Freq = 37 GHz (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.000	0.900	0.870	0.570
Ant dia- azimuth	1.40	1.00	1.50	1.50
Ant spin rate	15	50	100	115
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	850.00	200.00	100.00	60.00
Xmt pulse width	0.1500	0.0800	0.0600	0.0380
PRF	85,000	300,000	500,000	500,000
<Tfa>	40.00	12.00	12.00	5.50
Design capability: # scans on target = N	3	3	3	3
PCR	1.00	1.00	1.00	1.00
Detection to Doppler (Time/scan)*	1	1	1	1
Total goals score	Rank-2 (lower)	Rank-1 (highest)	Rank-3 (lower)	Rank-4 (lowest)

Achieved goals

Example: 37 GHz Optimizing for # Scans Target Tracked

Example 37 GHz optimizing # Scans Target Tracked (varying Grazing Angle)

Case-A

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	6.0	deg
Ant dia- elevation	1.50	meter
Ant dia- azimuth		meter
Ant spin rate	110	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	1200.00	W
Xmt pulse width	0.1500	microsec
PRF	125,000	Hz
<Tfa>	6.00	sec
Design capability: # scans on target = N	7	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Max # scans

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	1.50	0.10	0%
Ant Az dia		0.10	
Target tracking, # spins	7	10	57%
Ant Spin rate	110	30	47%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	1200	10.0	20%
Swath width		5.00E+04	
Total goals score			Rank-1

Example 37 GHz optimizing # Scans Target Tracked (varying Grazing Angle)

Case-B

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	12.0	deg
Ant dia- elevation	0.870	meter
Ant dia- azimuth		meter
Ant spin rate	178	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	1250.00	W
Xmt pulse width	0.1000	microsec
PRF	295,000	Hz
<Tfa>	12.00	sec
Design capability: # scans on target = N	6	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Max # scans

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.87	0.10	45%
Ant Az dia		0.10	
Target tracking, # spins	6	10	43%
Ant Spin rate	178	30	1%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	1250	10.0	17%
Swath width		5.00E+04	
Total goals score			Rank-2

Example 37 GHz optimizing # Scans Target Tracked (varying Grazing Angle)

Case-C

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	18.0	deg
Ant dia- elevation	0.570	meter
Ant dia- azimuth		meter
Ant spin rate	180	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	325.00	W
Xmt pulse width	0.0398	microsec
PRF	450,000	Hz
<Tfa>	3.50	sec
Design capability: # scans on target = N	5	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Max # scans

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.57	0.10	66%
Ant Az dia		0.10	
Target tracking, # spins	5	10	29%
Ant Spin rate	180	30	0%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	325	10.0	78%
Swath width		5.00E+04	
Total goals score			Rank-3

Example 37 GHz optimizing # Scans Target Tracked (varying Grazing Angle)

Case-D

Parameters not shown are constant

Radar Design Parameters Choices		
Radar Freq	3.70E+10	Hz
UAV Altitude		km
Ant.boresight grazing ang	24.0	deg
Ant dia- elevation	0.510	meter
Ant dia- azimuth		meter
Ant spin rate	180	rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr	150.00	W
Xmt pulse width	0.0380	microsec
PRF	500,000	Hz
<Tfa>	5.50	sec
Design capability: # scans on target = N	4	integer
PCR	1.00	
Detection to Doppler (Time/scan)*	1	ratio

Max # scans

Design compliant score			100%
Goals	Design	Goal	% of Goal
Ant El dia	0.51	0.10	71%
Ant Az dia		0.10	
Target tracking, # spins	4	10	14%
Ant Spin rate	180	30	0%
Warning time		60	
Doppler precision goal (worst-case)		50.0	
Transmit power	150	10.0	90%
Swath width		5.00E+04	
Total goals score			Rank-4

Example 37 GHz optimizing # Scans Target Tracked (varying Grazing Angle)

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0

Increasing Grazing Angle

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180

Increasing Spin Rate

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	1200.00	1250.00	325.00	150.00

Decreasing Pt

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	1200.00	1250.00	325.00	150.00
Xmt pulse width	0.1500	0.1000	0.0398	0.0380

Decreasing tau

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	1200.00	1250.00	325.00	150.00
Xmt pulse width	0.1500	0.1000	0.0398	0.0380
PRF	125,000	295,000	450,000	500,000

Increasing PRF

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	1200.00	1250.00	325.00	150.00
Xmt pulse width	0.1500	0.1000	0.0398	0.0380
PRF	125,000	295,000	450,000	500,000
<Tfa>	6.00	12.00	3.50	5.50

Tracking Spin Rate <Tfa>

Example 37 GHz optimizing # Scans Target Tracked

Parameters not shown are constant

Radar Design Parameters Choices	Case-A	Case-B	Case-C	Case-D
Radar Freq	3.70E+10	3.70E+10	3.70E+10	3.70E+10
UAV Altitude				
Ant.boresight grazing ang	6.0	12.0	18.0	24.0
Ant dia- elevation	1.50	0.870	0.570	0.510
Ant dia- azimuth				
Ant spin rate	110	178	180	180
Antenna Azimuth +/- from fore-look				
Xmtr Pwr	1200.00	1250.00	325.00	150.00
Xmt pulse width	0.1500	0.1000	0.0398	0.0380
PRF	125,000	295,000	450,000	500,000
<Tfa>	6.00	12.00	3.50	5.50
Design capability: # scans on target = N	7	6	5	4
PCR	1.00	1.00	1.00	1.00
Detection to Doppler (Time/scan)*	1	1	1	1

Decreasing Max # scans Target tracking

Submit your test cases to validate your model before you conduce trade studies!